National University of Lesotho

Department of Mathematics and Computer Science

CS4433 Software Engineering

Individual Final Project

Moderated Learning Management System

Verification and Validation

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1. Introduction

1.1 Purpose

This document shows how the Moderated LMS, version 1.0, was checked. It contains a test plan, test code, test results, a coverage report and a defect log. Table 1.1 shows where each one is.

Table 1.1: The deliverables of Phase 5

Deliverable	Where it is
Test plan	Section 2 of this document
Test code	The tests folder of the Git repository: the GoogleTest tests (tests/test_rules.cpp, tests/test_workflow.cpp, tests/test_security.cpp), the system tests (tests/system/run_system_tests.py), the scripted sessions (tests/e2e) and the timing test (tests/bench.cpp)
Test results	Section 3.2 (automated tests), Section 4.3 and 4.5 (system and acceptance tests)
Coverage report	Section 3.3
Defect log	Section 5

1.2 What was tested, and where

The tests ran on the source code of the Git repository at the end of Phase 5. Table 1.2 shows the tools and the computer that were used.

Table 1.2: Test environment

Item	Value
Operating system	Ubuntu Linux 24.04
Computer	2 processor cores, Intel Xeon 2.8 GHz
Compiler and build	g++ 13.3 (C++17) and CMake 3.28
Test tools	GoogleTest 1.14 (unit and integration tests), gcov (coverage), Python 3 (system tests)
Libraries of the program	SQLite 3.45.1 and libsodium 1.0.18
CI pipeline	GitHub Actions, the file .github/workflows/ci.yml

1.3 How to run the tests

All commands are run from the main folder of the repository, after the program is built (see README.md).

Table 1.3: Commands to run the tests

Command	What it runs
ctest --test-dir build --output-on-failure	The 36 unit and integration tests (about 1 second)
python3 tests/system/run_system_tests.py	The 95 system tests (about 2.5 minutes)
sh tests/coverage.sh	The coverage of the business layer
sh tests/e2e/run_demo_session.sh	One full session through all four roles
python3 tests/e2e/fuzz_menus.py	60 sessions of wrong input
build/tests/lms_bench	The timing test

2. Test strategy and test plan

2.1 The V-Model and the four test levels

In the V-Model every design artefact on the left side has a test level on the right side that checks it. Table 2.1 shows the four levels, the artefact that each level verifies, what is tested, and when a level may start (entry criteria) and when it is finished (exit criteria).

Table 2.1: Test levels mapped to the artefacts they verify

Level	Artefact it verifies	What is tested and how	Entry criteria	Exit criteria
Unit	Detailed design: the class diagram and the business rules BR-01 to BR-10 (Tables 8, 9 and 10 of the SRS)	One class or function at a time, without the database: dates, the late penalty, the status changes, which status a student may see, the permissions of the four roles, the validation rules. 9 GoogleTest tests, suite Rules.	The code compiles with no warning. The class or function is finished.	All unit tests pass. Every rule of Tables 8, 9 and 10 has a test. Line coverage of the business layer is at least 70% (NFR-10).
Integration	Architecture design: the layers, the interfaces in Ports.h, the repositories, the database schema and the sequence diagrams	The services working together with the real SQLite repositories (on an in-memory database), a fake clock and a fast fake password hasher: submit, resubmit, extension, marking, moderation, release, log in and lock, access control, SQL injection, audit log, rollback. 27 GoogleTest tests, suites Workflow (16) and Security (11).	All unit tests pass.	All integration tests pass. A failed action leaves the data unchanged.
System	The SRS: the functional requirements FR-01 to FR-26, the non-functional requirements, and the use case specifications (UC-01 to UC-20)	The whole program is started like a user would start it, with typed input. Each test follows the main flow, an alternative flow or an exception flow of a use case. The text on the screen and the data in the database are both checked. 95 tests ST-01 to ST-95, the scripted session, the wrong-input test and the timing test.	All integration tests pass. The program is built from a clean copy of the repository.	Every Must requirement has at least one passing system test. Every system test passes. No defect of severity High or Medium is open.
Acceptance	The needs of the stakeholders (Principal, Heads of Department, lecturers, students, IT officer) behind the requirements	Whole scenarios that each stakeholder asked for, run through the real program (Section 4.3).	All system tests pass.	Every scenario passes. No defect is open.

Rule for a failed test. When a test fails, a defect is written in the defect log (Section 5). The defect is fixed, then the test that failed and all the earlier tests are run again (retest and regression test). A level is finished only when its exit criteria are met.

2.2 Test design: equivalence partitioning

For each input, the possible values are split into classes. All values in one class should behave in the same way, so one value from each class is tested. Table 2.2 shows the classes.

Table 2.2: Equivalence classes

Input	Valid classes	Invalid classes	Tests
Raw mark (the maximum mark is 50)	0 to 50, with at most one decimal place	Below 0; above 50; not a number; two decimal places	Rules.validation_rules; ST-54, ST-58, ST-59, ST-60, ST-61
Maximum mark of an assignment	A whole number from 1 to 100	0 or less; 101 or more; not a number	Rules.validation_rules; ST-29, ST-30
Submitted file	From 1 byte to 5 MB	File not found; empty file; bigger than 5 MB	Workflow.file_checks_empty_too_big_and_not_enrolled; ST-42 to ST-45
Time of a submission (compared with the deadline of the student)	On time; 1 to 24 hours late; more than 24 and up to 48; more than 48 and up to 72	More than 72 hours late (accepted, but the mark is 0)	Rules.late_penalty_bands_and_boundaries; ST-41, ST-46, ST-47, ST-57
Password	At least 8 characters, with a letter and a digit	Under 8 characters; no digit; no letter	Rules.validation_rules; ST-06, ST-07
Date and time text	A real date and time written as 2026-11-10 12:00	Words; month 13; 30 February; 29 February in a year that is not a leap year; hour 24; minute 60; extra text	Rules.datetime_rejects_wrong_input; ST-31
Status of a submission when it is changed	The 7 changes of Table 9	The other 29 changes	Rules.status_changes_follow_table_9; ST-50, ST-51, ST-66, ST-81
Role of the user for a function	The functions that Table 10 gives to the role	Every other function; another lecturer's course; another department	Security.every_service_function_is_refused_to_the_wrong_role; Security.own_course_and_own_department_checks; ST-33, ST-63, ST-75

2.3 Test design: boundary value analysis

Mistakes are most often made at the edge of a class, so the values on the edge and next to it are tested. Table 2.3 shows them.

Table 2.3: Boundary values

Item	**Boundary values tested**	**Expected result**	**Tests**
Late penalty (BR-01)	Exactly at the deadline; 1 minute late; exactly 24 hours; 24 hours and 1 minute; exactly 48 hours; 48 hours and 1 minute; exactly 72 hours; 72 hours and 1 minute; 10 days	Days late 0, 1, 1, 2, 2, 3, 3, 4, 10; penalty 0, 5, 5, 10, 10, 15, 15, 40, 40 (maximum mark 50, raw mark 40)	Rules.late_penalty_bands_and_boundaries; Workflow.submission_exactly_at_the_deadline_is_on_time
Penalty bigger than the raw mark	Raw mark 10 and penalty 15; raw mark 5 and penalty 10 (1 day late, maximum 100)	The final mark is 0, never below 0	Rules.late_penalty_never_gives_a_negative_mark; ST-57
Maximum mark of an assignment	0, 1, 100, 101 (and 200)	1 and 100 accepted; 0, 101 and 200 refused	Rules.validation_rules; ST-29; ST-30

Item	Boundary values tested	Expected result	Tests
Raw mark (maximum 50)	-1, 0, 50, 50.1, 51	0 and 50 accepted; the others refused	Rules.validation_rules; ST-58; ST-59
Size of the file	0 bytes, 5,242,880 bytes (exactly 5 MB), 5,242,881 bytes	Only the first is refused as empty; 5 MB accepted; 5 MB plus 1 byte refused	Workflow.file_checks_empty_too_big_and_not_enrolled; ST-43; ST-44; ST-45
Wrong passwords in a row (FR-03)	4 wrong, the 5th wrong, 4 minutes later, 5 minutes later	Locked after the 5th; still locked after 4 minutes; open after 5 minutes	Security.five_wrong_passwords_lock_the_account_for_five_minutes; ST-04
New deadline of an extension (BR-04)	Earlier than the present deadline; the same as the present deadline; later	Only a later deadline is accepted	Workflow.extension_rules; ST-35
Leap year	29 February 2028 at 23:59 plus 1 minute	1 March 2028 00:00	Rules.datetime_parses_and_prints

2.4 Limits of the plan

- The exact times of the late penalty (24, 48 and 72 hours) cannot be set with the real clock of the computer. They are tested at unit level with a fake clock (Table 2.3). The system tests use real times that are far from the boundaries (2 days late, more than 3 days late, 20 hours late).
- The tests ran on one Ubuntu computer. They were not run on Windows.
- The times in the requirements (NFR-01, NFR-02) are for the reference computer. The tests ran on another computer, so the times are a guide only.
- NFR-07 (five people try the system with the user manual) and NFR-11 (a technician installs the system in 30 minutes) need people. They belong to Phase 7, when the user manual and the installation guide exist.

3. Automated unit and integration tests

3.1 The tests

The automated tests are written with GoogleTest. There are 36 tests in three suites. Rules has 9 unit tests that need no database. Workflow (16) and Security (11) are integration tests: they use the real services, the real SQLite repositories and the real schema on an in-memory database, with a fake clock so that times can be set exactly. Table 3.1 lists all 36 tests.

Table 3.1: The 36 GoogleTest tests

Suite	Test	Level	What it checks	Requirement
Rules	datetime_parses_and_prints	Unit	Reading and printing dates; adding minutes; a leap year	FR-09
Rules	datetime_rejects_wrong_input	Unit	Eight kinds of wrong date text are refused	FR-09
Rules	late_penalty_bands_and_boundaries	Unit	The bands of Table 8 with the boundary times (10 cases)	FR-13, BR-01
Rules	late_penalty_never_gives_a_negative_mark	Unit	A penalty bigger than the raw mark gives the final mark 0	FR-13, BR-01
Rules	late_penalty_keeps_one_decimal_without_rounding_error	Unit	37.3 with maximum 33, 2 days late: penalty 6.6, final 30.7, exactly	FR-13, BR-01
Rules	status_changes_follow_table_9	Unit	Exactly 7 of the 36 status changes are allowed; a wrong change is refused	FR-24, BR-06
Rules	students_see_only_submitted_or_late_until_release	Unit	The status a student may see before the release	FR-19, BR-02
Rules	roles_own_only_the_functions_of_table_10	Unit	The functions of the four roles	FR-02, BR-10
Rules	validation_rules	Unit	Password, mark, maximum mark, text and username rules	BR-08, BR-09
Workflow	on_time_submission_has_status_submitted_and_a_receipt	Integration	An on-time submission gets the status Submitted and a receipt	FR-16
Workflow	submission_exactly_at_the_deadline_is_on_time	Integration	A submission at the deadline minute is on time	FR-16, BR-01

Suite	Test	Level	What it checks	Requirement
Workflow	late_work_is_accepted_with_a_warning_and_cannot_be_replaced	Integration	Late work is accepted, the penalty is shown, it cannot be replaced	FR-18, BR-03
Workflow	work_more_than_72_hours_late_is_accepted_but_gets_zero	Integration	Work more than 72 hours late is accepted and gets 0	FR-18
Workflow	resubmission_before_the_deadline_keeps_the_older_versions	Integration	A new version is saved; the old versions are kept	FR-17
Workflow	resubmission_after_the_deadline_or_after_marking_is_refused	Integration	No replacement after the deadline or after marking	FR-17, BR-03
Workflow	file_checks_empty_too_big_and_not_enrolled	Integration	Empty file, 5 MB plus 1 byte, 5 MB accepted, no file name, unknown assignment, student not enrolled	FR-16
Workflow	extension_makes_late_work_on_time_again	Integration	An extension changes Late back to Submitted	FR-10, BR-04
Workflow	extension_recalculates_the_days_late_when_still_late	Integration	An extension that is still too early gives new days late	FR-10, BR-04
Workflow	extension_rules	Integration	Earlier or equal deadline, no reason, other lecturer, marked work, second extension	FR-10, BR-04
Workflow	marking_applies_the_penalty_and_writes_one_audit_entry	Integration	Marking saves the penalty and exactly one audit entry	FR-12, FR-13, FR-14
Workflow	lecturer_can_change_a_mark_only_with_a_reason_and_only_while_marked_or_returned	Integration	A change needs a reason and the right status	BR-06

Suite	Test	Level	What it checks	Requirement
Workflow	moderation_approve_adjust_return_and_release	Integration	Approve, change, send back and release, with the audit log and messages	FR-20 to FR-23
Workflow	students_see_no_mark_before_release_and_the_mark_after_release	Integration	A student sees the mark only after the release	FR-19, BR-02
Workflow	student_assignment_list_shows_the_deadline_of_the_student_and_not_submitted	Integration	The list shows the own deadline and the status	FR-15
Workflow	administrator_rules_for_users_departments_and_courses	Integration	Users, departments, courses, lecturers and enrolment	FR-04 to FR-08
Security	login_works_and_does_not_tell_which_part_was_wrong	Integration	Log in works; one message for a wrong user or password	FR-01
Security	five_wrong_passwords_lock_the_account_for_five_minutes	Integration	Lock after the 5th wrong password; open after 5 minutes	FR-03
Security	a_good_login_resets_the_count_of_wrong_passwords	Integration	The count is of wrong passwords in a row	FR-03
Security	administrator_can_open_a_locked_account_with_a_new_password	Integration	A new password opens a locked account	FR-05
Security	every_service_function_is_refused_to_the_wrong_role	Integration	Every service function with every wrong role	FR-02, NFR-04
Security	own_course_and_own_department_checks	Integration	Own course, own department and own work only	FR-02, BR-07
Security	sql_injection_does_not_work	Integration	SQL text typed as input is only data	AR-3
Security	the_audit_log_cannot_be_changed_or_deleted	Integration	No change or delete of an audit entry	NFR-09

Suite	Test	Level	What it checks	Requirement
Security	a_failed_action_leaves_the_data_unchanged	Integration	A rollback; a failure in the middle of saving a mark	NFR-08, AR-5
Security	database_rules_back_up_the_services	Integration	The schema refuses wrong data, as a second barrier	FR-04, FR-07
Security	passwords_are_stored_only_as_salted_argon2id_hashes	Integration	Argon2id hash, own salt, 100 to 500 ms	NFR-03

3.2 Results

All 36 tests pass. This is the end of the output of the test program:

```
[==========] 36 tests from 3 test suites ran.
[  PASSED  ] 36 tests.
```

CTest also reports each test one by one: **100% tests passed, 0 tests failed out of 36**. The whole run takes about 1 second. The test Security.passwords_are_stored_only_as_salted_argon2id_hashes prints the time of one Argon2id check: 127 ms on the test computer (NFR-03 asks for 100 to 500 ms).

3.3 Coverage report

The brief asks for at least 70% line coverage of the business-logic layer (NFR-10) and for the actual figure. The coverage was measured with gcov on the files src/business/*.cpp, using only the 36 automated tests. The command is sh tests/coverage.sh. Table 3.2 shows the result.

Table 3.2: Line coverage of the business layer

File	Lines covered	Lines covered / lines
AdminService.cpp	68.3%	138 of 202
AssignmentService.cpp	68.4%	67 of 98
AuditService.cpp	100%	8 of 8
AuthService.cpp	100%	38 of 38
DateTime.cpp	90.9%	50 of 55
Entities.cpp	97.5%	39 of 40
MarkingService.cpp	100%	82 of 82
MessageService.cpp	88.9%	8 of 9
ModerationService.cpp	97.8%	134 of 137
PenaltyStrategy.cpp	100%	11 of 11
SubmissionService.cpp	97.5%	115 of 118
Types.cpp	66.7%	44 of 66
User.cpp	85.2%	23 of 27
UserFactory.cpp	80.0%	12 of 15
Validation.cpp	100%	49 of 49
Total business layer	**85.7%**	**818 of 955**

The actual figure is 85.7% (818 of 955 lines). This is above the target of 70%.

Three files are below 70%. The lines that the automated tests do not reach are these:

- AdminService.cpp: editing a user, editing a department, editing a course, and the refusal to deactivate a department that still has a Head of Department.
- AssignmentService.cpp: the three functions that only list data for the menus (listMine, listMyCourses, listEnrolledStudents).
- Types.cpp: the text names of roles and statuses for the screen, and the error branches for unknown values.

These parts are not left untested. The system tests run them through the real program (for example ST-11 edits a user, ST-17 edits a department and a course, ST-19 deactivates a department, ST-34 lists the students for an extension). The coverage figure above counts only the automated unit and integration tests.

3.4 The CI pipeline

The file .github/workflows/ci.yml defines the pipeline of GitHub Actions. On every push and every pull request it does these steps: install the tools, build the program, run the 36 GoogleTest tests, run the 95 system tests, run the scripted session and the wrong-input test, and measure the coverage. The coverage step stops the pipeline with an error if the coverage is under 70%.

Status. The commands of every step were run by hand on the test computer, and all of them pass. The pipeline itself has not run on GitHub yet, because the repository is not on GitHub yet. The first run will be shown in Phase 6 as the CI evidence.

4. System and acceptance testing

4.1 How the system tests were designed

The system tests come from the use case specifications in the SRS: UC-16 Submit assignment (Table 14), UC-08 Mark submission (Table 15) and UC-12 Review and moderate marks (Table 16), and from the requirements of the other use cases. For each use case there is a test for the main flow, a test for each alternative flow and a test for each exception flow. Tests were also made for the business rules and the non-functional requirements.

Each test follows the same steps: (1) start the real program on a new database that holds the demo data, (2) type the input that a user would type, (3) check the text that the program shows, (4) check what is saved in the database (for example the status, the number of versions, the audit entries). A test fails if the program stops with an error, writes to the error output, shows "Unexpected error", or prints a colour code (NFR-12).

The exception flow "database or disk error" (UC-16 E4, UC-08 E5, UC-12 E4) is checked in five ways: two in the integration tests (a transaction that is not committed is rolled back; a failure in the middle of saving a mark leaves nothing behind) and three at start-up, with a file that is not a database (ST-92), a database of this program with another version number (ST-93), and a database of another program (ST-95).

Table 4.1: System tests for each use case

Use case	Name	Number of tests	Result
UC-01	Log in	6	Pass
UC-02	Manage users	10	Pass
UC-03	Manage departments and courses	6	Pass
UC-04	Assign lecturer to course	2	Pass
UC-05	Enrol students	4	Pass
UC-06	Create assignment	6	Pass
UC-07	Grant extension	4	Pass
UC-08	Mark submission	9	Pass
UC-09	Write feedback	1	Pass
UC-10	Calculate late penalty	3	Pass
UC-11	Record audit entry	4 (inside UC-08 and UC-13: ST-54, ST-64, ST-69, ST-89)	Pass
UC-12	Review and moderate marks	8	Pass
UC-13	Adjust mark	4	Pass
UC-14	Release marks	4	Pass
UC-15	View courses and assignments	3	Pass
UC-16	Submit assignment	7	Pass
UC-17	Submit late assignment	3	Pass
UC-18	Resubmit assignment	3	Pass
UC-19	View released marks and feedback	3	Pass
UC-20	View audit log	2	Pass
NFR	Non-functional requirements (NFR-03, NFR-05, NFR-08, NFR-09)	7	Pass

4.2 The system test cases

Table 4.2 lists all 95 system tests with the expected result and the result of the final run. The typed input of each test is in tests/system/run_system_tests.py.

Table 4.2: System test cases and results (final run)

ID	Use case	What is tested	Expected result	Result
ST-01	UC-01	Log in with each of the four roles	Each user sees a welcome line with the right role.	Pass
ST-02	UC-01	Wrong password (exception flow)	The error 'Wrong username or password.' is shown.	Pass
ST-03	UC-01	Unknown username gives the same message	The same error as for a wrong password.	Pass
ST-04	UC-01	Five wrong passwords lock the account (FR-03)	After the 5th wrong password the account is locked; even the right password is then refused.	Pass
ST-05	UC-01	Empty username and password	The program asks for both and logs nobody in.	Pass
ST-06	UC-02	Create a student; the new student can log in; the password is stored as a hash	User created, the log in works, and the database holds an Argon2id hash and no plain password.	Pass
ST-07	UC-02	Password that is too weak (BR-08)	An error is shown and no user is created.	Pass
ST-08	UC-02	Username already used	An error is shown and the second account is not created.	Pass
ST-09	UC-02	The two passwords are not the same	An error is shown and no user is created.	Pass
ST-10	UC-02	Second active Head of Department in the same department (FR-04)	An error is shown and the second Head is not created.	Pass
ST-11	UC-02	Edit the name of a user	The change is confirmed and the new name is in the user list.	Pass

ID	Use case	What is tested	Expected result	Result
ST-12	UC-02	Deactivate a user; that user cannot log in any more	The user is deactivated and the next log in is refused.	Pass
ST-13	UC-02	The Administrator cannot deactivate the own account	An error is shown.	Pass
ST-14	UC-02	Set a new password (FR-05)	The new password works and the old one does not.	Pass
ST-15	UC-02	A new password opens a locked account	After 5 wrong passwords and a new password, the user logs in.	Pass
ST-16	UC-03	Create a department and a course	Both are created.	Pass
ST-17	UC-03	Edit a department and a course	Both changes are confirmed.	Pass
ST-18	UC-03	Course code already used	An error is shown and no course is created.	Pass
ST-19	UC-03	Deactivate a department that still has a course and a Head	An error is shown.	Pass
ST-20	UC-03	Deactivate a course	The course is deactivated.	Pass
ST-21	UC-03	Empty department name	An error is shown.	Pass
ST-22	UC-04	Assign a lecturer to a course	The assignment is confirmed and saved.	Pass
ST-23	UC-04	Assign a student as a lecturer (exception)	An error is shown.	Pass
ST-24	UC-05	Enrol a student	The student is enrolled.	Pass
ST-25	UC-05	Remove a student from a course	The student is removed.	Pass
ST-26	UC-05	Enrol a lecturer as a student (exception)	An error is shown.	Pass
ST-27	UC-05	Enrol the same student twice (exception)	An error is shown.	Pass
ST-28	UC-06	Create an assignment	The assignment is created and is in the list.	Pass
ST-29	UC-06	Maximum mark outside 1 to 100 (boundary values 101, 0, 200)	After 3 wrong entries the action is cancelled.	Pass
ST-30	UC-06	Maximum marks 1 and 100 (boundary values)	Both are accepted.	Pass
ST-31	UC-06	Wrong date formats	After 3 wrong entries the action is cancelled.	Pass
ST-32	UC-06	Empty title	An error is shown.	Pass
ST-33	UC-06	Course that the lecturer does not teach (FR-02)	An error is shown.	Pass
ST-34	UC-07	Extension for a student; the student sees the new deadline and is on time	The extension is saved; the student list shows the new deadline; a submission is now on time.	Pass
ST-35	UC-07	New deadline earlier than the present one (exception)	An error is shown and nothing is saved.	Pass
ST-36	UC-07	Extension without a reason (exception)	An error is shown and nothing is saved.	Pass
ST-37	UC-07	Extension after a late submission makes it on time again (BR-04)	The status goes from Late back to Submitted and the days late go back to 0.	Pass
ST-38	UC-15	A student sees only the assignments of the own courses	student1 (enrolled in MA2101) sees it; student4 (not enrolled) does not.	Pass
ST-39	UC-15	A student who is not enrolled does not see MA2101	The course code is not in the list of student4.	Pass
ST-40	UC-15	Status 'Not submitted' before and 'Submitted' after submitting	The status text changes.	Pass
ST-41	UC-16	Submit on time (main flow)	A receipt with version 1 and the status Submitted.	Pass
ST-42	UC-16	File does not exist (E1)	An error is shown and nothing is saved.	Pass
ST-43	UC-16	File is empty (E1)	An error is shown and nothing is saved.	Pass
ST-44	UC-16	File of 5 MB plus 1 byte (E1, boundary)	An error is shown and nothing is saved.	Pass
ST-45	UC-16	File of exactly 5 MB (boundary)	The file is accepted.	Pass
ST-46	UC-17	Submit 2 days late (A1)	A warning with the penalty, then status Late with 2 days.	Pass
ST-47	UC-17	Submit more than 72 hours late (A2)	A warning that the mark will be 0; the work is saved as Late.	Pass
ST-48	UC-17	The student cancels a late submission (A1)	Nothing is saved.	Pass
ST-49	UC-18	Resubmit before the deadline (A3)	Version 2 is saved and version 1 is kept.	Pass
ST-50	UC-18	Late work cannot be replaced (E2, BR-03)	An error is shown and still one version.	Pass
ST-51	UC-18	Marked work cannot be replaced (E2, BR-03)	An error is shown and still one version.	Pass
ST-52	UC-16	Student not enrolled in the course (E3)	An error is shown and nothing is saved.	Pass

ID	Use case	What is tested	Expected result	Result
ST-53	UC-16	Assignment does not exist (E3)	An error is shown.	Pass
ST-54	UC-08	Mark an on-time submission (main flow)	Raw 40, penalty 0, final 40; status Marked; one audit entry 'initial marking'.	Pass
ST-55	UC-10	Mark a submission that is 2 days late (BR-01)	Penalty 4.0 (2 x 10% of 20); final 11.0.	Pass
ST-56	UC-10	Mark work more than 72 hours late (A3)	The penalty is the whole raw mark; final 0.	Pass
ST-57	UC-10	The penalty is bigger than the raw mark: final mark is 0, not negative (boundary)	Raw 5, 1 day late, maximum 100: penalty 10.0, final 0.0.	Pass
ST-58	UC-08	Raw mark above the maximum (E1)	An error is shown and nothing is saved.	Pass
ST-59	UC-08	Raw mark 0 and raw mark 50 of 50 (boundary values)	Both are accepted.	Pass
ST-60	UC-08	Raw mark is not a number (E1)	After 3 wrong entries the action is cancelled.	Pass
ST-61	UC-08	Raw mark with two decimals (BR-09)	An error is shown and nothing is saved.	Pass
ST-62	UC-09	Empty feedback (E2)	An error is shown and nothing is saved.	Pass
ST-63	UC-08	A lecturer who does not teach the course (E3)	An error is shown and no submission list.	Pass
ST-64	UC-08	Change a mark before moderation needs a reason (A2)	With a reason: saved with a second audit entry.	Pass
ST-65	UC-08	Change a mark without a reason (E2)	An error is shown; still one audit entry.	Pass
ST-66	UC-08	A Moderated mark can no longer be changed (E4, BR-06)	An error is shown; the status stays Moderated.	Pass
ST-67	UC-12	Approve a mark (main flow)	The status becomes Moderated.	Pass
ST-68	UC-12	The Head sees the raw mark, penalty, final mark and feedback (FR-20)	All are shown.	Pass
ST-69	UC-13	Change the final mark (A1)	The change is saved and a second audit entry has old 40, new 35 and the reason.	Pass
ST-70	UC-13	New final mark above the maximum (E3)	An error is shown; still one audit entry.	Pass
ST-71	UC-13	Change the final mark without a reason (E3)	An error is shown; still one audit entry.	Pass
ST-72	UC-13	Final mark 0 is accepted (boundary)	The change is saved with new final mark 0.	Pass
ST-73	UC-12	Send back to the lecturer (A2); the lecturer sees a message and marks again	Status Returned; the lecturer gets a message at the next log in; the new mark is saved with a reason.	Pass
ST-74	UC-12	Send back without a comment (E3)	An error is shown; the status stays Marked.	Pass
ST-75	UC-12	Head of another department (E1, BR-07)	hod_maths sees no work and cannot approve it.	Pass
ST-76	UC-12	Nothing waiting for moderation (E2)	The message 'There are no marked submissions...' is shown.	Pass
ST-77	UC-12	Assignment summary (FR-25)	The summary lists the assignment and the counts.	Pass
ST-78	UC-14	Release one moderated submission	The mark is released; the status is Released.	Pass
ST-79	UC-14	Release all moderated submissions of an assignment	Two submissions are released.	Pass
ST-80	UC-14	A submission that is only Marked cannot be released (FR-22)	Nothing is released.	Pass
ST-81	UC-14	A Released mark cannot be changed (BR-06)	An error is shown; the final mark stays.	Pass
ST-82	UC-19	The student sees no mark before the release (BR-02, FR-19)	No mark and no feedback; the status shows Submitted.	Pass
ST-83	UC-19	The student sees the mark and feedback after the release	The final mark, the feedback and the message at log in are shown.	Pass
ST-84	UC-19	Another student does not see the released mark (NFR-04)	student2 sees no mark.	Pass
ST-85	UC-12	The lecturer gets a message when the mark is released (FR-23)	The message is shown at the next log in.	Pass
ST-86	UC-20	The Head sees the audit log of the department	The entry with 'initial marking' is shown.	Pass
ST-87	UC-20	The Head of another department sees none of it (BR-07)	The log is empty for hod_maths.	Pass
ST-88	NFR-03	Every stored password is a salted Argon2id hash	All hashes start with $argon2id and the six students (same password) have six different hashes.	Pass
ST-89	NFR-09	The audit log cannot be changed or deleted	The database refuses UPDATE and DELETE on audit_entry.	Pass

ID	Use case	What is tested	Expected result	Result
ST-90	NFR-08	Data stays after the program is closed and started again	A user made in the first run can log in the second run.	Pass
ST-91	UC-01	First start without demo data: the first Administrator is created	The program asks for the first Administrator; no default password exists.	Pass
ST-92	NFR-08	A file that is not a database (database error)	The program says that it cannot start and stops with exit code 1; no crash.	Pass
ST-93	NFR-08	A database of this program with another version number is refused	The program says that it cannot start; exit code 1.	Pass
ST-94	NFR-05	The source code has no network code (works offline)	No socket, HTTP or network header is used in src/.	Pass
ST-95	NFR-08	A database of some other program is refused and not changed	The program says that it cannot start; the file still has only its own table.	Pass

4.3 Acceptance tests

The acceptance tests check the needs of the stakeholders, not the details of a requirement. Each scenario is run through the real program from the first step to the last. The stakeholder is the source that the SRS names for the requirement (Table 5 of the SRS).

Table 4.3: Acceptance tests

ID	Stakeholder	Need and scenario	Evidence	Result
AT-01	Principal	Students must not see a mark until the Head of Department releases it. A student submits, the lecturer marks, the Head approves and releases, and only then the student sees the final mark and the feedback.	tests/e2e/run_demo_session.sh (14 checks); ST-82, ST-83, ST-84	Pass
AT-02	Student representative	Late work is accepted, and the student is told the penalty before the file is saved.	ST-46, ST-47, ST-48	Pass
AT-03	Student representative	A student can replace the work before the deadline, and the old file is kept.	ST-49	Pass
AT-04	Head of Department	The Head sees the raw mark, the penalty and the final mark; can change a final mark with a reason that goes in the audit log; can send work back with a comment.	ST-68, ST-69, ST-73, ST-86	Pass
AT-05	Lecturer	A lecturer gives one student a later deadline, and the student then submits on time.	ST-34, ST-37	Pass
AT-06	Lecturer and student	A lecturer or student sees a message at the next log in when work is sent back or released.	ST-73, ST-83, ST-85	Pass
AT-07	IT officer	The system works without a network, stores passwords safely and is fast enough.	ST-94, ST-88; timing test (Table 4.4)	Pass

Two acceptance tests of the SRS need people and are not done yet: NFR-07 (five people try a task with the user manual) and NFR-11 (a technician installs the system in 30 minutes). They belong to Phase 7.

4.4 Results of the non-functional checks

Table 4.4: Non-functional checks

NFR	Check	Result	Verdict
NFR-01	30 log ins with 1,008 users stored (tests/bench.cpp)	Average 119.7 ms; slowest 140.6 ms; 0 of 30 over 2 seconds	Pass*
NFR-02	Save a 5 MB file; save a mark; list 800 submissions (20,000 stored)	20.8 ms; 0.2 ms; 2.4 ms (limit 2,000 ms each)	Pass*
NFR-03	Passwords in the database; time of one check	All 11 stored hashes start with $argon2id (ST-88); the 6 students with the same password have 6 different hashes; one check takes 127 ms	Pass
NFR-04	Every function with every wrong role	Security.every_service_function_is_refused_to_the_wrong_role; ST-33, ST-63, ST-75, ST-84	Pass
NFR-05	Network code in the source	None found in src/ (ST-94). The program was not run with the network switched off.	Pass (source check only)
NFR-06	Steps after log in (counted in the system tests)	Submit: 4 (menu choice, assignment, file, confirm). Mark: 6 (7 for a change). Approve: 3. Release all: 3; release one: 4. The limit is 7.	Pass
NFR-08	Wrong input and database problems	60 sessions of random and wrong input (letters, huge numbers, quotes, bad dates, missing files, 5,000 characters): 0 problems. Five database problems: see Section 4.1.	Pass
NFR-09	Audit log	The database refuses UPDATE and DELETE on audit_entry (ST-89)	Pass
NFR-12	Text only, no colour	No colour code in the output of any of the 95 tests; every function works with the keyboard	Pass
NFR-13	1,008 users, 102 courses, 20,000 submissions	The timing test above was made with this data	Pass*

* The times were measured on the test computer, not on the reference computer. They are far under the limits, but they must be measured again on the reference computer before they are used as proof.

4.5 Results of the test runs

The system tests were run three times. After the last fix they were also run on the old code, to prove that the failures were real defects.

Table 4.5: Runs of the system tests

Run	Situation	Tests	Passed	Failed	Failed tests
Run 1	First run	94	89	5	ST-18, ST-21, ST-82, ST-84, ST-93
Run 2	After the fixes of DEF-03 to DEF-05 and the corrections of the test script; ST-95 was added	95	94	1	ST-93
Run 3	Final run (after the last correction of the test script)	95	95	0	None
Check	The final tests on the code before the fixes	95	92	3	ST-18, ST-21, ST-95

Some failures were mistakes in the test script and not defects of the program (Section 5.2).

5. Defect log

5.1 Defects of the program

The severity is High (wrong marks, lost data, a security failure or a crash), Medium (a requirement or rule is not met, or data of another program is at risk) or Low (a wrong or misleading text, no data is wrong). Every defect has been fixed and tested again. No defect is open.

Table 5.1: Defect log

ID	Found by	Description	Severity	Cause	Fix	Retest
DEF-01	Phase 4: the test Security.passwords_are_stored_only_as_salted_argon2id_hashes, which prints the time	One password check took 94 ms. NFR-03 asks for 100 to 500 ms.	Medium	The Argon2id setting was too light.	The number of passes was raised to 4 (64 MiB of memory).	Pass. One check takes about 127 ms.
DEF-02	Phase 4: looking at the menu screens when running the program	In the menu tables, a long text ran into the next column.	Low	The function that prints a cell did not cut a long text or leave a space.	format::cell cuts a long text with a ~ and always leaves a space.	Pass. The tables read well in the scripted session.
DEF-03	Phase 5: ST-18 and ST-21	After a failed action the screen showed "Course created. The id is" or "Department created. The id is" and then the error. No data was saved, but the text was wrong.	Low	The text and the call of the service were in one output statement, so the text was printed before the call failed.	The service is called first and the text is printed after it. (AdministratorMenu::createCourse and createDepartment)	Pass. ST-18 and ST-21 pass.
DEF-04	Phase 5: code review after DEF-03	The same pattern was in the extension screen of the lecturer. No test could show a wrong output there.	Low	The same as DEF-03.	The same fix. (LecturerMenu::grantExtension)	Pass. ST-34 to ST-37 pass.
DEF-05	Phase 5: ST-95, written with ST-93	The program accepted a database file of another program and added 34 entries (tables, indexes and triggers) to it.	Medium	The check of the version number ran only when the table user_account existed. Any other file was treated as a new file.	A file that has tables but no user_account table is refused and not changed. (Database::ensureSchema)	Pass. ST-95 passes and the file keeps its 1 table; ST-93 passes.
DEF-06	Phase 5: the build for coverage (Debug, with gcov)	The compiler warned -Wformat-truncation in DateTime::toString.	Low	The text buffer of 32 bytes was too small in the worst case that the compiler can see.	The buffer was made 64 bytes.	Pass. The build has 0 warnings in Debug and in Release.

5.2 Mistakes in the test script

These failures were not defects of the program. They were found and corrected in the test script.

- ST-82 and ST-84 (Run 1): the test checked the whole output, but the screen of the Head of Department also shows the feedback. The check now looks only at the screen of the student.
- ST-93 (Run 1 and Run 2): the first version of the test made a database without the tables of the program. That test showed DEF-05, so a new test ST-95 was added for it, and ST-93 now uses a real database of the program with the version number changed.

6. Summary

Table 6.1: Results at a glance

Item	Result
Unit and integration tests (GoogleTest)	36 of 36 pass
Line coverage of the business layer	85.7% (818 of 955 lines); the target is 70%
System tests	95 of 95 pass (after the fixes)
Acceptance tests	7 of 7 pass; 2 need people and belong to Phase 7 (NFR-07, NFR-11)
Scripted session through all four roles	14 of 14 checks pass
Wrong-input test	60 sessions, 0 problems
Defects	6 logged, 6 fixed and tested again, 0 open

What is not done or not proved: the tests were not run on Windows; the times are from the test computer and not from the reference computer; the program was not run with the network switched off (only the source was checked); the CI pipeline has not run on GitHub yet; NFR-07 and NFR-11 need people.